Advanced Programming Practice · 21CSC203P

Unit 1 · Session 11 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 11 : SLO – 2 | Subroutine Overhead & Memory

Activity (Solutions)

• 1) Call-stack trace of recursion: fact(3) → pushes frame fact(3); calls fact(2) → frame fact(2); calls fact(1) → frame fact(1) returns 1; fact(2) returns 2*1=2; fact(3) returns 3*2=6 — each call pushes a new stack frame (parameters, locals, return address); returns pop them. Depth too large → StackOverflowError.

• 2) Track memory: Runtime rt = Runtime.getRuntime(); long used = rt.totalMemory() - rt.freeMemory(); System.out.println("Used: " + used/1024 + " KB"); — measure before/after allocations.

• 3) Static vs dynamic creation: a static/class-level object is created once (loaded with the class) → small fixed footprint, shared by all instances; dynamic creation with new inside a loop allocates a fresh heap object per iteration → footprint grows with count until GC runs. E.g., 10000 new StringBuilder() vs one reused instance — the static/reused one uses far less memory.